Yuto Abe

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Summary

ML engineer bridging research and production: built and published a 7B-parameter Japanese full-duplex spoken dialogue model (first author, IWSDS 2026), built its evaluation infrastructure (Full-Duplex-Bench-JA; Spoken Dialogue Arena, oral at the RTCA workshop @ NeurIPS 2026), and optimized Elasticsearch analytics over hundreds of millions of records in production. 2.5 years of industry experience spanning ML infrastructure, full-stack web development, and large-scale data processing.

Education

Waseda University — Tokyo, Japan
M.S. in Computer Science & Communications — *Apr. 2025 – Mar. 2027 (Expected)*
- Research: Full-duplex spoken dialogue systems, MLOps for speech models
- Advisers: Prof. Tetsunori Kobayashi, Prof. Tetsuji Ogawa
- Member of LLM-jp Dialogue Working Group (National Institute of Informatics)

Waseda University — Tokyo, Japan
B.S. in Information & Communication Engineering (GPA: 3.33/4.00) — *Apr. 2021 – Mar. 2025*

Work Experience

ABEJA, Inc. — Tokyo, Japan
Forward Deployed Engineer Intern — *Mar. 2026 – Present*
- Building internal product management applications using Next.js and FastAPI with authentication
- Onboarding to ABEJA's ML platform products for enterprise AI deployment

3-shake Inc. — Tokyo, Japan
SRE Workshop — *Oct. 2026 (2 days)*
- Hands-on SRE practice: container and Kubernetes fundamentals, GitOps deployment automation, monitoring, and reliability design

TIER IV, Inc. — Tokyo, Japan
Autoware Workshop 2026 — MLOps & AI Track — *Aug. – Sep. 2026*
- Developed an end-to-end driving model using the Autoware autonomous-driving OS and the AWSIM simulator

GMO Internet Group — Tokyo, Japan
ML/Web Engineering Intern — *Feb. 2026 (2 weeks)*
- Implemented end-to-end MVP of "Tenbin AI" (AI-powered comparison tool) during intensive internship

National Institute of Informatics — LLM-jp Project — Tokyo, Japan
Research Engineer — *Jun. 2025 – Present*
- First-authored **LLM-jp-Moshi-v1**, a 7B-parameter Japanese full-duplex spoken dialogue model, outperforming the J-Moshi baseline in naturalness and semantic appropriateness
- Designed and built evaluation infrastructure for full-duplex spoken dialogue (**Full-Duplex-Bench-JA**, VAP-distance) and co-operate the human A/B evaluation platform **Spoken Dialogue Arena**
- Built Apache Airflow DAGs for the training pipeline on ABCI HPC (PBS Pro): a custom PBSJobOperator, a 100-way fan-out TTS synthesis pipeline (about 46K wavs, 530 hours), and a 7-stage training DAG with retry-on-preemption
- Trained on Japanese dialogue corpora (J-CHAT, CSJ, CallHome-Japanese) with curriculum learning, using DeepSpeed + Accelerate (mixed precision, ZeRO Stage 2) on multi-GPU ABCI clusters

Medley, Inc. — Tokyo, Japan
Engineering Intern — *Oct. 2025*
- Improved LLM-based medical dialogue summarization that transcribes doctor-patient conversations via ASR
- Resolved drug name omission, relative date expression, and context-dependent reference issues by injecting structured medical context into prompts, enabling accurate clinical summaries

Hitachi, Ltd. — Tokyo, Japan
Research Intern — *Aug. – Sep. 2025*
- Conducted fake audio detection experiments using Qwen-Audio, a large-scale audio-language model

Accenture Tokyo, Japan
Engineering Intern *Aug. 2025*
- Built a warehouse management application as part of a consulting engineering program

Japan Useware Systems Co., Ltd. Tokyo, Japan
Software Engineer (Part-time) *Feb. 2023 – Oct. 2025*
- Designed and implemented an access log analytics dashboard processing **hundreds of millions of records** with Elasticsearch aggregation and stacked bar chart visualization (React + MUI)
- Resolved API response delays and browser freezes by optimizing query windows to 400 days, tuning aggregation bucket counts, and capping frontend rendering at 400 data points — **reduced response time by 10x**

Publications & Presentations

Yuto Abe et al.
- "Spoken Dialogue Arena: A Live, Full-Duplex A/B Evaluation Platform for Spoken Dialogue Systems." **RTCA @ NeurIPS 2026** (Real-Time Conversational Agents Workshop). *First author, oral.*
- "Full-Duplex-Bench-JA: Building a Benchmark for Evaluating Japanese Full-Duplex Spoken Dialogue Models." **ASJ Autumn Meeting 2026** (Acoustical Society of Japan). *First author, oral.*

Yuto Abe, M. Saeki, A. Ohashi, S. Fujie, T. Kobayashi, et al.
- "Effects of Dialogue Corpora Properties on Fine-Tuning a Moshi-Based Spoken Dialogue Model." **IWSDS 2026** (International Workshop on Spoken Dialogue Systems). *First author.*

Yuto Abe et al.
- Presentation at **ASJ Spring Meeting 2026** (Acoustical Society of Japan). *First author.*

Projects

Spoken Dialogue Arena — Human Evaluation Platform for Spoken Dialogue Models | *Live* | *GitHub*
- Built a real-time A/B evaluation platform enabling blind comparison of full-duplex speech dialogue models, addressing the lack of standardized human evaluation tools for conversational AI
- Designed scalable architecture for **100+ concurrent users**: Next.js (App Router), PostgreSQL, Prisma ORM, Cloudflare Tunnel, GPU inference servers with WebRTC streaming
- Implemented Bradley-Terry ranking integration for statistically grounded model comparison

Global Mirai Parliament & Political Quiz App — Team Mirai (Civic Tech)
- Lead student engineer for civic-tech initiative bridging technology and policy-making
- Developed "Global Mirai Parliament" (adapts overseas policies for Japanese context) and a political quiz app for voter engagement

Awards

OpenAI Codex Student Builder Fest 2026 — Top 4 2026
- Built "Kotoba Link": connects speech, lip reading, text, and sign language with AI to make conversation possible in online meetings

Progate Hackathon — Grand Prize ("Samether.io") 2026

YANS 2026 Hackathon — Excellence Award (Original Model Division) 2026
- Hallucination-detection task at the 21st YANS symposium; identified the generator's perturbation patterns through data analysis

PKSHA Hackathon — Grand Prize (1st Place) 2025
- Built "Changemakers": AI-powered medical consultation and appointment booking app using GPT-Realtime API, enabling patients to complete hospital intake interviews via real-time voice interaction

PR TIMES Hackathon — Individual Excellence Award ("Dream Blossom", a goal-sharing web app) 2025

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, Java, Scala, PHP, Bash
ML/AI: PyTorch, Hugging Face Transformers, DeepSpeed, Accelerate, ESPnet, MLflow, Qwen-Audio, Mimi codec
Web: React, Next.js, Node.js, FastAPI, Hono, Prisma, Supabase
Databases: PostgreSQL, MySQL, Elasticsearch, DynamoDB
Infrastructure: Docker, Kubernetes, GCP, AWS, Cloudflare, WebRTC/WebSocket, GitHub Actions
ML Infrastructure: Apache Airflow, multi-GPU distributed training (ABCI HPC), NCCL, mixed precision, ZeRO